

Computer System

- 1 Describe the differences between ROM and RAM and explain what they are used for.

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..... [3]

(June 2006)

- 2 Tick whether the following uses are more suitable for ROM or RAM.

	ROM	RAM
Storing data that the user is currently working on		
Storing the system BIOS		
Storing program instructions for computer games		

[3]

(June 2007)

- 3 Tick the box which best describes one of the following items.

	Hardware	Software
Word processor		
Video camera		
Scanner		
Web page design program		

[4]

(Nov 2004)

4 Give three features of a graphical user interface.

Feature 1 _____

Feature 2 _____

Feature 3 _____

Feature 4 _____ [4]

(Specimen 2007)

5 Tick **True** or **False** for the following statements.

	True	False
ROM is volatile.		
ROM is used to store the startup instructions of a computer.		
The data in ROM is easier to change than that in RAM.		
ROM is used to store the data the user is currently working on.		

[4]

(Nov 12 V.2)

6 Describe **three** drawbacks of using a command line interface (CLI) operating system.

1

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2

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..... [3]

(Jun 13 V.1)

- 7 Give **three** advantages of using a graphical user interface (GUI) rather than a command line interface (CLI).

1

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2

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..... [3]

(Jun 13 V.2)

- 8 Describe **four** features of ROM.

1

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2

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3

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4

..... [4]

(Jun 13 V.2)

- 9 Tick **True** or **False** next to each of these statements.

	True	False
Computer programs are examples of software.		
A web authoring package is an example of hardware.		
An icon is a component of a command line interface.		

[3]

(Nov 12 V.1)

10 Define the term Hardware. Name its two types and give three examples for each type.

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..... [4]

11 Discuss three advantages and three disadvantages of using tablets compared to using laptops.

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..... [6]

Input Devices

1 Name the input devices **A**, **B**, **C** and **D** using the words from the list.

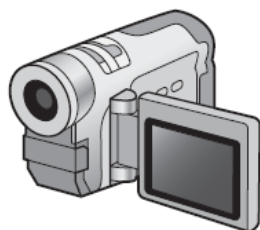
A



B



C



D



keyboard

microphone

remote control

scanner

sensor

touch pad

touch screen

video camera

A

B

C

D

[4]

(Jun 13 V.2)

2 Ring **two** items which are input devices.

Buzzer

Dot matrix printer

Graph plotter

Graphics tablet

Laser printer

Touch screen

[2]

(June 2009)

- 3 Draw **five** lines on the diagram to match the input device to its most **appropriate** use.

Input Device	Use
Mouse	to input voiceovers for a presentation
Joystick	to input a PIN
Optical Mark Reader	to select options from a menu
Microphone	to operate a flight simulator
Number pad	to input candidate responses on examination papers

[5]

(Nov 12 V.2)

- 4 Draw **four** lines on the diagram to match the use to its most **appropriate** input method.

Use	Input method
To input text from a printed document	Bar code reader
To input data from a bank cheque	Chip reader
To input data from a bank card	MICR
To input data about a product at a POS	OCR

[4]

(Jun 11 V.2)

- 5 Ring **two** items which are input devices.

DVD RAM	laser printer	magnetic disc
number pad	speakers	trackerball

[2]

(Jun 13 V.1)

- [6]

[6]

8 Name three applications in which RFID technology is used.

1

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2

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3

..... [3]

9 A customer uses a bank card to purchase a product at a supermarket checkout. The checkout operator uses a number of devices to input data about the customer and the product.

Identify **four** of these input devices and for each one describe a different item of data that will be input.

Device 1

Data

.....

Device 2

Data

.....

Device 3

Data

.....

Device 4

Data

..... [8]

(Jun 13 V.1)

Output Devices

- 1 Ring **two** items which are output devices.

DVD ROM

Hard disc

Keyboard

Plotter

Robot arm

Scanner

(June 2007)

- 2 Tick **True** or **False** next to each of these statements.

	True	False
Dot matrix printers produce high quality output.		
Laser printers are very noisy.		
Graph plotters are used when extremely large hard copy is required.		
Inkjet printers are used where continuous stationery is required.		

[4]

(Nov. 2009)

- 3 Ring **two** items which are output devices.

Blu-ray disc

Graph plotter

Graphics tablet

Optical mark reader

Projector

Web cam

[2]

(Jun 10 V.1)

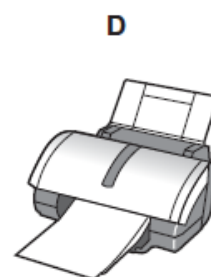
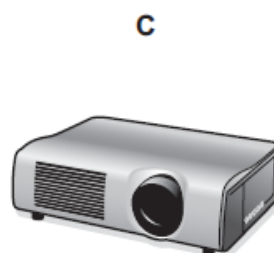
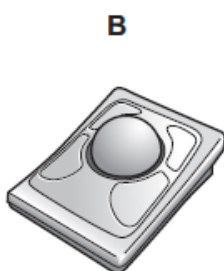
- 4 Draw **four** lines on the diagram to match the use to its most **appropriate** output device.

Use	Output device
to produce very large high quality hard copy	dot matrix printer
to display data from a computer on to a large screen	plotter
to alert mobile phone users to incoming calls	multimedia projector
to output multipart forms	buzzer

[4]

(Jun 11 V.3)

- 5 Name the devices **A**, **B**, **C** and **D** using the words from the list.



bar code

dot matrix printer

flash memory card

inkjet printer

monitor

multimedia projector

scanner

trackerball

A

B

C

D

[4]

(Nov 12 V.2)

6 Give three advantages of using LED monitor over the traditional LCD monitor.

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..... [3]

7 Explain what is meant by 3-D printers, giving two advantages and two disadvantages of using them.

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..... [6]

Storage Devices

- 1 Complete the following sentences using the most suitable storage device or medium listed below.

DVD R

fixed hard disk drive

memory stick

- (a) A student taking his coursework home to develop would use a

..... [1]

- (b) A company selling copies of games would use a

..... [1]

- (c) An office worker who never took her work home would store her work on a

..... [1]

(June 2007)

- 2 Ring **two** storage devices.

CD ROM drive

Memory stick

Mouse

Speaker

Temperature sensor

Trackerball

[2]

(Nov 2007)

- 3 Draw **five** lines on the diagram to match the storage device to its most appropriate use.

Storage Device	Use
Fixed hard disc	Making backups of a file server
Flash memory card	Transferring a file from one computer to another
Magnetic tape	Storing software sold by a company
Pen drive	Storing data which needs to be accessed quickly
DVD ROM	Storing photographs in a digital camera

[5]

(Nov 11 V.2)

- 4 Complete each sentence below using one item from the list.

bar code reader

chip

DVD RW

keyboard

laser printer

magnetic tape

- (a) The **most suitable** storage medium for making backups of data held on a network server is a
- (b) The part of a bank card that stores data about the PIN is a
- (c) An optical disc which can have data updated is a
- (d) Details of a product are scanned at a POS using a

[4]

(Nov 11 V.3)

- 5 Ring **two** items which are used for data storage.

light pen

magnetic disc

motor

mouse

OCR

ROM

[2]

(Nov 12 V.2)

6 Tick **True** or **False** next to each of these statements.

	True	False
Computer programs are examples of software.		
A web authoring package is an example of hardware.		
An icon is a component of a command line interface.		
A mobile phone (cell phone) uses flash memory.		

[4]

(Nov 12 V.1)

7 Use the **most appropriate** phrase from the list below to complete each sentence.

store data in real time applications

store high definition films

store backups of file servers

store music for sale

store photographs in a digital camera

- (i) A CD ROM is used to
- (ii) A blu-ray disc is used to
- (iii) A magnetic tape is used to
- (iv) A flash memory card is used to

[4]

(Jun 13 V.2)

- 8 Susan uses a memory stick to transfer her work from school to home. Give **three** reasons why memory sticks are often preferred to CDs for storing work.

.....

.....

.....

.....

..... [3]

(Nov 11 V.1)

- 9 Describe **two** uses of optical media and for each one give a reason why optical media are preferred to magnetic tapes for that use.

Use 1

.....

Reason 1

.....

Use 2

.....

Reason 2

..... [4]

(Nov 11 V.3)

10 Solid State Drives (SSDs) are rapidly taking over from HDDs.

(a) Give three advantages of using SSDs over HDDs.

- 1
-
- 2
-
- 3
-

[3]

(b) Explain why SSDs are more suitable than HDDs for laptops.

-
-
-
-
-
-
-

[3]

11 Instead of saving data on a local hard disk or other storage device, a user can save their data 'on the cloud'.

(a) Explain what is meant by Cloud Storage.

-
-
-
-

[2]

(b) Name and describe the security risks associated with using Cloud Storage.

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..... [4]

(c) Name three advantages of using Cloud Storage.

1.....

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3

..... [3]

Introduction to databases

1 Select the most suitable data type from the list to represent:

- (a) the number of children in a family
- (b) a line of a company's address
- (c) the gender of a person
- (d) somebody's birthday

Alphanumeric

Boolean

Date

Numeric (Integer)

Numeric (Real)

[4]

(June 2009)

2 A school keeps records of every student including details of their names, addresses, phone numbers and dates of birth. Using this information, define the terms file, field and record.

File

Field

Record

[3]

(June 11 V.1)

- 3 A school keeps records of its students, using many different types of data. Explain what is meant by Boolean, alphanumeric and numeric data, using examples of student data.

(a) Boolean

Meaning

.....

Example

.....

(b) Alphanumeric

Meaning

.....

Example

.....

(c) Numeric

Meaning

.....

Example

.....

[6]

(Jun 11 V.3)

- 4 The manager of a bookshop uses a database to store data about the books she sells. This is part of the database.

Title of book	Fiction (Y/N)	Number in stock	Next order due
Full Circle	N	15	20/06/11
Sanditon	Y	18	08/07/11
Vanity Fair	Y	23	16/10/11
The Old Curiosity Shop	Y	10	23/06/11
Pride and Prejudice	Y	12	23/06/11
Sense and Sensibility	Y	26	02/10/11
The Moon's a Balloon	N	12	21/08/11

- (a) How many fields are there in this part of the database?

..... [1]

- (b) How many records are there in this part of the database?

..... [1]

- (c) The records shown are to be sorted in ascending order of **Number in stock**. What will be the title of the book which will be the first record in the database after it has been sorted?

..... [1]

- (d) What is the title of the book which has no more than 12 copies in stock and is not Fiction?

..... [1]

(Jun 11 V.3)

5 Describe the differences between validation and verification giving **two** examples of validation checks.

[5]

(June 2008)

- 6 A school library has a file for storing details of the books it has and a file for storing details of its borrowers. The two files are linked using a common field.

Book File

Code	Title	Author	Published	Number in stock	Cost
1857028898	The Code Book	Simon Singh	1999	2	£10.99
0747591054	The Deathly Hallows	J K Rowling	2007	8	£17.99
0748791167	Sepulchre	Kate Mosse	2007	3	£18.99
0563371218	Full Circle	Michael Palin	1997	1	£19.99

Years 10 and 11 Borrower File

Number	Name	Form	Book Borrowed	Due Back
0102	Me Te Loan	11A	1857028898	14/06/2010
1097	Gurvinder Moore	10C	0747591054	12/06/2010
0767	Akhtar Aftab	10B	0748791167	06/06/2010
0611	Graham Reeves	10D	0563371218	08/06/2010

- (a) How many records are there in the book file?

..... [1]

- (b) How many fields are there in the borrower file?

..... [1]

- (c) What type of database do these two files form?

..... [1]

- (d) Give **two** reasons why this type of database system is used rather than having two flat files.

1

.....

2

..... [2]

(e) Which field is the primary key in the Book file?

..... [1]

(f) Which field is the foreign key?

..... [1]

(g) Which field has the data type currency?

..... [1]

(h) What data type would be most appropriate for the **Due Back** field?

..... [1]

(Jun 10 V.1)

7

Make and model	Number sold	Air conditioning (Y/N)
Ford Mondeo	45	Y
Opel Corsa	52	N
BMW Series X	35	N
Nissan Micra	39	N

A car showroom uses a database to store information about the cars it has sold. Part of this database is shown above.

(a) (i) Which field stores Boolean data?

.....

(ii) Give one reason why this data type is used.

.....

..... [2]

(b) (i) Which field stores numeric data?

.....

(ii) Give one reason why this data type is used.

.....

..... [2]

(Jun 11 V.1)

8 Chi has employed Sarbjit, a systems analyst, to create a new database system for his mobile telephone (cellphone) business.

- He keeps a number of different models in stock.
- Most of the phones have a camera but some do not.
- The rental plans have 100, 200 or 500 free texts depending on the amount the customer pays per month.
- The minimum monthly payment is \$10 and the maximum is \$100.

(a) Complete the data dictionary below filling in the field names and the **most appropriate** data types to create a database using only the above information.

Field name	Data type
Model	
Monthly payment	

[6]

(b) Name and describe **three** different validation checks that could be used on the data in this database.

1

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.....

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2

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3

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..... [6]

(Jun 13 V.2)

- 9 The manager of a travel company asked her secretary to look into the reliability of trains in Birmingham. The secretary obtained the details of some of the trains. He created two spreadsheets to help him do this.

Sheet 1 is a list of some of the UK stations that trains from Birmingham go to. Sheet 2 consists of the train details he looked at.

	A	B
1	Code	Station
2	E	Edinburgh
3	LB	Longbridge
4	B	Bournemouth
5	LI	Lichfield
6	S	Shrewsbury
7	W	Wolverhampton
8	LO	London
9	P	Plymouth

Sheet 1

(Commas are used as delimiters in the functions shown below.)

	A	B	C	D
1	Arrival Station	Code	Number	
2	=VLOOKUP(B2,Sheet1!\$A\$2:\$B\$9,2)	E	=COUNTIF(\$C\$9:\$C\$18,B2)	
3	=VLOOKUP(B3,Sheet1!\$A\$2:\$B\$9,2)	LO	=COUNTIF(\$C\$9:\$C\$18,B3)	
4	=VLOOKUP(B4,Sheet1!\$A\$2:\$B\$9,2)	S	=COUNTIF(\$C\$9:\$C\$18,B4)	
5	=VLOOKUP(B5,Sheet1!\$A\$2:\$B\$9,2)	LI	=COUNTIF(\$C\$9:\$C\$18,B5)	
6	=VLOOKUP(B6,Sheet1!\$A\$2:\$B\$9,2)	W	=COUNTIF(\$C\$9:\$C\$18,B6)	
7				
8	Arrival time	Minutes late	Arrival code	Late Y/N
9	11:03	0	E	=IF(B9>0,"Y","N")
10	11:05	4	LO	=IF(B10>0,"Y","N")
11	11:07	-4	W	=IF(B11>0,"Y","N")
12	11:10	13	LO	=IF(B12>0,"Y","N")
13	11:12	6	S	=IF(B13>0,"Y","N")
14	11:14	-2	LI	=IF(B14>0,"Y","N")
15	11:16	0	W	=IF(B15>0,"Y","N")
16	11:19	-3	LO	=IF(B16>0,"Y","N")
17	11:22	0	LI	=IF(B17>0,"Y","N")
18	11:25	6	E	=IF(B18>0,"Y","N")

Sheet 2

(a) Explain what the function in cell A2 in Sheet 2 does.

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..... [3]

(b) What station would you expect to see in cell A3 in Sheet 2?

..... [1]

(c) Explain what the function in cell C2 in Sheet 2 does.

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..... [3]

(d) What value would you expect to see in cell C4 in Sheet 2?

..... [1]

(e) Explain what the function in cell D9 in Sheet 2 does.

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..... [3]

(f) What value would you expect to see in cell D11 in Sheet 2?

..... [1]

- 10 Iqbal wants to test the new computerised payroll system he would like to introduce to his company. No company worker is paid less than \$100 and no worker is paid more than \$500.

Explain what is meant by the following three types of test data using examples of the wages paid to workers.

Normal

.....

.....

.....

Abnormal

.....

.....

.....

Extreme

.....

.....

..... [6]

(Jun 13 V.1)

Networks

- 1 Give **two** advantages and **two** disadvantages of having computers networked.

Advantage 1

.....

Advantage 2

.....

Disadvantage 1

.....

Disadvantage 2

..... [4]

(June 2008)

- 2 A large company has a LAN. The manager of the company wants to replace it with a WLAN. Explain the differences between a LAN and a WLAN, giving an advantage and a disadvantage of a WLAN.

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..... [4]

(Jun 11 V.1)

3 A desktop computer is already used in an office which does not have an internet connection.

(a) Name **two** items of hardware which could enable the computer to be connected to a network.

1

2 [2]

(b) Name **one** item of hardware which would enable the network to be connected to the internet.

..... [1]

(Jun 11 V.1)

4 Aftab and his family have three computers in their home. He wants to connect the computers into a network. Explain why he would need:

A router

.....

A browser

.....

Email

.....

An ISP

..... [4]

(Jun 10 V.1)

5 Tick **three** items of hardware which are used to network computers together.

	☒
Hub	☐
Printer	☐
Scanner	☐
Switch	☐
Bridge	☐
Monitor	☐

[3]

(Nov 11 V.2)

6 Describe the routing process done by the router.

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[4]

- 7 Explain why switches are more intelligent devices than hubs.

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..... [3]

- 8 Tick **Internet** or **intranet**, next to each statement, as appropriate.

	Internet	intranet
is a network of computer networks		
exists usually within one organisation		
anybody can access it		
can be expanded to become an extranet		

[4]
(Nov 13 V.1)

- 9 Explain what is meant by the following terms.

Website

.....

Webserver

.....

Web browser

..... [3]

10 Explain what is meant by the following.

HTTP

.....

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HTTPS

.....

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FTP

.....

..... [3]

11 Explain the difference between the internet and World Wide Web.

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.....

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..... [3]

12 Describe **three** advantages of electronic faxing over physical faxing.

1.....
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2
.....

3
..... [3]

13 The Regionalisa Bank sends messages to its customers. Sometimes the messages are sent by fax and sometimes by email. Tick whether the following statements are **TRUE** or **FALSE**.

	TRUE	FALSE
A faxed message is more private		
An emailed message arrives with the customer more quickly		
A faxed document can be used for legal purposes		
An email attachment can be edited electronically		

[4]

(Jun 11 V.2)

14 (a) Explain what is meant by VoIP.

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.....
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..... [2]

(b) Describe **two** advantages and **two** disadvantages of using this technology.

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..... [4]

15 As well as PCs, mobile phones (cellphones) can be used to access the Internet.

(a) Give **two** advantages of using a PC rather than a mobile phone for this purpose.

1

.....

2

..... [2]

(b) Give **two** disadvantages of using a PC rather than a mobile phone for this purpose.

1

.....

2

..... [2]

(Nov 13 V.1)

Extract the following from the URL:

<http://www.google.com.eg/googlesearch/pages/home.htm>.

Protocol

Domain Name

Domain Type

Country Code

Filename

Path [6]

17 Encrypted data is used in on-line banking. Describe encryption.

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.....

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..... [3]

(Jun 11 V.3)

18 Explain the difference between strong password and weak password, giving an example of each.

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..... [4]

19 A bank is looking into a variety of security measures to protect its customers' data. Describe **three** types of authentication technique they could use and for each give a benefit and a drawback.

Technique 1

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20 Explain what is meant by the following terms.

Phishing

.....

Pharming

.....

Spam

..... [3]

(Jun 10 V.1)

21 (a) Name two possible effects caused by spam.

1.....

.....

2

..... [2]

(b) Describe four measures to reduce the risk of spam.

1.....

.....

2

.....

3

.....

4

..... [4]

22 The internet is not owned by anyone and thus, not policed.

(a) List three arguments in favor of having some control over the internet.

- 1
-
- 2
-
- 3
- [3]

(b) List three arguments against the idea of having control over the internet.

- 1
-
- 2
-
- 3
- [3]

23 Describe how it is possible to be the subject of fraud when using a credit card online.

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.....

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.....

.....

..... [4]

System Development Life Cycle

- 1 Mohammed Hussain owns a bookshop whose sales have grown enormously. He now wants to have a computer system to take over the running of the shop. He has employed a systems analyst to advise him on the type of system he needs. Before she can recommend a new system she will need to analyse the current system.

- (a) Describe **four** methods of researching/analysing and for each one write down an advantage of using that method.

Method 1

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.....

.....

Method 2

.....

.....

.....

Method 3

.....

.....

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Method 4

.....

.....

..... [8]

- (b) The systems analyst decides that the system needs to have a screen input form to enter all the details of each book. Design a screen input form which could be used for typing in the details of **one** book.



[5]

- (c) The systems analyst now designs the file structure. Give **three** items which would need to be designed as part of the file structure.

1

.....

2

.....

3

.....

[3]

(Nov 2008)

- 2 The new system will be tested. The month part of the date of birth field must be a whole number between 1 and 12 inclusive. Using this as an example, describe the following three types of test data.

(i) Abnormal data

.....

.....

..... [2]

(ii) Normal data

.....

.....

..... [2]

(iii) Extreme data

.....

.....

..... [2]

(Nov 10 V.1)

- 3 After the system is implemented it will be evaluated. Identify **three** activities which would need to be carried out in evaluating the system.

1

.....

2

.....

3

..... [3]

(Nov 2008)

- 4 After a system is designed it will be implemented. Describe each of the following methods of implementation.

(i) Direct changeover

.....

.....

(ii) Parallel running

.....

.....

(iii) Phased implementation

.....

.....

(iv) Pilot running

.....

..... [4]

(Nov 11 V.1)

- 5 A test is marked out of 20 and recorded as an integer. Tick whether each of the following marks is an example of **abnormal** data or **extreme** data.

	Abnormal	Extreme
20		
21		
twenty		
0		

[4]

(Jun 12 V.1)

- 6 After the system is implemented the travel agent will be given documentation to go with it. This will consist of technical documentation and user documentation. Explain the purpose of each type of documentation and **two** different items that each type of documentation will contain.

Technical documentation

Purpose

.....

Item 1

.....

Item 2

.....

User documentation

Purpose

.....

Item 1

.....

Item 2

..... [6]

(June 2008)

- 7 (a) Name four factors that should be considered when planning and creating ICT solutions.

1

2

3

4 [4]

(b) Describe how data can be collected about the target audience.

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..... [3]

Effects of using ICT

- 1 Copyright law applies whenever you buy software. Give **three** actions that would break copyright law.

1

.....

2

.....

3

..... [3]

(June 2008)

- 2 The prolonged use of computers can lead to health and safety problems.

(a) Tick **two** health problems that may be caused by prolonged use of computers.

RSI	☐
Influenza virus	☐
Blood poisoning	☐
Headaches	☐

[2]

(b) Tick **two** ways of preventing health problems.

Sit far away from the monitor	☐
Sit in a comfortable chair	☐
Take regular breaks	☐
Put a screen filter in front of the monitor	☐

(c) Tick **two** safety problems that may be caused by the use of computers.

Back problems	☐
Electrocution	☐
Fire	☐
Hacking	☐

[2]

(d) Tick **two** ways of preventing safety problems.

Don't overload electrical sockets	☐
Make sure there are no bare wires	☐
Use an expert system	☐
Use anti-virus software	☐

(June 2009)

[2]

3 Tick **three** developments which have arisen because of the internet.

	☒
Web logs	☐
Databases	☐
Models	☐
Wikis	☐
DTP	☐
Social networking sites	☐

(Jun 10 V.1)

4 Tick whether the following statements apply to **Blogs** or apply to **Wikis**.

	Blogs	Wikis
Usually in some form of chronological order	☐	☐
Usually many contributors and authors	☐	☐
Usually personal	☐	☐
Postings tend to be short in length	☐	☐

(Jun 12 V.2)

- 5 The use of computers can lead to safety problems. Describe **two** safety problems and how they could be prevented.

Problem 1

.....

Prevention

.....

Problem 2

.....

Prevention

..... [4]

(June 2008)

- 6 Computers can cause health problems. Identify **two** of these problems and for each problem write down a method which would help you avoid it. Do not use the same method for both answers.

Problem 1

.....

Method 1

.....

Problem 2

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Method 2

..... [4]

(Jun 11 V.1)

- 7 There are many microprocessor-controlled devices in the modern home. Describe the effects of these devices on people's lifestyles.

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..... [4]

(Jun 11 V.2)

- 8 Explain what is meant by the following.

Part-time working

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Flexible hours

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Job sharing

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Compressed hours

..... [4]

9 Explain what is meant by personal data, giving three examples.

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..... [2]

10 (a) Explain what is meant by Data Protection Acts (DPAs).

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..... [2]

(b) List four principles of Data Protection Acts.

1.....

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2

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3

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4

..... [4]

- 11 (a) Describe the differences between physical safety and e-safety when using computers.

.....
.....
.....
..... [2]

- (b) Describe **three** measures which should be taken in order to maintain e-safety.

1
.....
2
.....
3
..... [3]

(Specimen 2016)

- 12 Describe four methods that software producers employ to prevent software copyright being broken.

1.....
.....
2
.....
3
.....
4
..... [4]

ICT Applications

- 1 Tick **four** items which must be part of an expert system.

Bar code reader	☐
Spreadsheet	☐
Dot matrix printer	☐
Inference engine	☐
Interactive input screen	☐
Knowledge base	☐
Rules base	☐
Search engine	☐

[4]

(June 2009)

- 2 A group of doctors want to have an expert system to help them with their diagnoses. Describe how such a system would be created and evaluated.

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[5]

(June 2008)

3 A car repair centre uses an expert system to help diagnose car engine faults.

(a) Describe the inputs, outputs and processing of this system.

[6]

(b) Give **two** other examples of situations where expert systems might be used.

1

2

[2]

(Nov 2009)

4 An automatic washing machine is controlled by a microprocessor.

(a) Tick **two** sensors which would have to be used in the machine.

	☒
Pressure sensor	☐
Proximity sensor	☐
Temperature sensor	☐
Oxygen sensor	☐

[2]

(b) Explain why computers are unable to read the data directly from these sensors.

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.....

.....

..... [2]

(c) Describe how the microprocessor uses data from the sensors to control the washing machine.

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..... [5]

(Nov 2009)

- 5 Give **three** reasons why pilots are trained using flight simulators rather than by flying real aircraft.

1

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2

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3

..... [3]

(Jun 10 V.1)

- 6 A hospital uses computers to monitor the health of patients.

(a) Give **three** advantages of using computers rather than nurses to do this.

1

.....

2

.....

3

..... [3]

(June 2009)

- 7 Microprocessors are used in different applications to control the process or to simply gather data (measurement only).

Tick which of the following applications are examples of control or of measurement only.

	Control	Measurement only
Automatic cookers		
Weather stations		
Microwave ovens		
Automatic washing machines		

(Jun 13 V.2)

- 8 Tick **three** advantages of using computers, rather than students, to control the growing conditions.

	☒
Computer readings are more accurate	☐
Students always forget to take readings	☐
Students might be unavailable to take readings during school holidays	☐
Computers can record data for later use	☐
Computers can analyse the results	☐
Computers can react to changes in the conditions immediately	☐

[3]

(Jun 10 V.1)

- 9 Describe **three** types of medical aid which can be produced using a 3D printer.

1

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2

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3

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[3]

(Specimen 2016)

- 10** OMR, OCR and MICR are methods of direct data entry.
Describe each method, giving an example of an application where each might be used.

(a) OMR
.....
.....
.....
.....
..... [3]

(b) OCR
.....
.....
.....
..... [3]

(c) MICR
.....
.....
.....
..... [3]

(Nov 11 V.1)

11 Describe how automatic number plate recognition (ANPR) systems work.

[6]

12 Explain what is meant by cookies, giving its use.

..... [3]

13 (a) Explain what is meant by GPS.

.....

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..... [3]

(b) Give **two** examples of the use of GPS.

1

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2

..... [2]

(Specimen 2016)

14 Explain what is AI biometrics used for. Give an example.

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..... [3]

15 Define the term Quantum cryptography, stating why it is used.

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..... [2]

- 16** Define the term Holography, naming three applications that make use of holographic images.

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..... [4]

- 17** There are areas outside manufacturing where robotics is evolving rapidly. Discuss.

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..... [4]

Banking applications

- 1 Tick three applications which use batch processing.

Producing utility bills.	
Monitoring a patient's condition in intensive care.	
Inputting details of new stock into a library's database.	
Robotics on a car production line.	
Controlling a greenhouse.	
Reading data from bank cheques.	

[3]

(June 2008)

- 2 Put a tick in the column which best describes the type of processing used in the following applications.

	On-line	Batch
Booking a holiday.		
Producing electricity bills.		
Using an ATM.		
Producing monthly payrolls.		

[4]

(Nov 2008)

- 3 Put a tick in the column which best describes the type of processing used in the following applications.

	Real Time	Batch
A burglar alarm system.		
Producing gas bills.		
Cooking with an automatic cooker.		
Scanning OMR sheets.		

[4]

(June 2009)

- 4 A bank uses a chip and pin system at its ATMs. A customer wishing to withdraw cash inserts their card. The ATM checks to see if the card is valid and if so asks the customer to type their PIN. Put the following steps in order using the numbers 2 to 8 for a customer whose transaction is successful. Step 1 has already been done for you.

The customer types in the PIN	1
If they are the same the customer is asked which service is required	
The customer's account is checked to see if it has sufficient funds	
The amount is checked against the card limit	
The typed PIN is compared with the PIN stored in the chip	
The customer selects required service (cash)	
If there are sufficient funds and the amount is within the card limit the required notes are issued.	
The customer is asked how much money they want to withdraw	

[7]

(Jun 10 V.3)

- 5 A supermarket uses POS terminals at its checkouts. This enables stock levels to be updated automatically. Put the following steps in order using the numbers 2 to 8. Step 1 has already been done for you.

The number in stock of the matching record is read	
Until a match is found with the entered bar code	
The data file is searched	
The bar code on the product is read by the bar code reader	1
The new value of number in stock is written back to the file	
If it is equal to the re-order number then more goods are automatically re-ordered	
One is subtracted from the number in stock	
The number in stock is compared with the re-order number	

[7]

(Jun 10 V.1)

- 6 Describe the batch processing of bank cheques.

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[4]

(Jun 11 V.1)

7 A supermarket has a number of EFTPOS terminals.

Explain what is meant by EFTPOS and how such a system works.

[6]

(Jun 13 V.2)

- 8 Suppose that John uses Hodder Bank and he pays a cheque for \$10,000 to company called H&S Ltd who banks with Smith Bank. Describe the process of clearing the cheque.

[1]